

Adding Within 10 — Finding and Fixing a Counting-On Mistake Answer Key

Kindergarten–Grade 1

Quick Answers

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|------|---------|----------|
| 1. 9 | 3. 8 | 5. 9 |
| 2. 6 | 4. 4, 4 | 6. 10, 8 |
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Curriculum

Level: Kindergarten–Grade 1

K.OA.A / 1.OA.C.6 – *problems 1, 2, 3, 4, 5, 6*

Difficulty 1–4 of 5 across 8 tagged checks.

Common wrong answers

1. If they got 8: said the first number as the first count instead of starting one number after it, so the total came out one too small
 2. If they got 5: counted the starting number as one of the hops, so one hop was missing at the end
 3. If they got 7: started at the smaller number and had so many hops to keep track of that one of them was lost
 4. If they got 5: counted the number names said out loud instead of the hops taken, and the starting number is said without any hop
 5. If they got 8: began the count at the small number and lost one of the seven hops along the way
 6. If they got 9: said the first number as hop one, which leaves the count one short of the real total
 6. If they got 7: made the same start-number slip a second time, so this total is one too small as well
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1. Find and fix: Sam, 6 stickers and 3 more.

Step 1: name the mistake. Sam said “six” as his first hop. But the six stickers are already there before he moves — saying the number he starts from costs no hop at all, so he only took two real hops instead of three.

Step 2: count on the right way. Hold 6 in your head, then hop: seven (one), eight (two), nine (three).

Step 3: Sam has 9 stickers.

Step 4: check it a different way. Count every dot in the picture from the beginning: six filled dots and then three more open ones is nine dots ✓.

Watch for: 8. A count-on answer that is exactly one too small is nearly always this mistake.

2. Find and fix: Mia, 4 blocks and 2 more.

(a) Step 1: Mia is adding two blocks, so she should take 2 hops. She said “four, five” — only one of those was a real hop, because “four” is the number she started from.

(b) Step 2: hold 4 in your head and hop twice: five (one), six (two).

Step 3: Mia has 6 blocks.

Step 4: check with fingers — hold up four fingers, then put up two more and count them all: six ✓.

Watch for: 5. Again exactly one too small.

3. Find and fix: Ravi, 3 cars and 5 more.

(a) Step 1: Ravi started at 3, the smaller number, which left him five hops to keep track of. Starting at 5, the bigger number, leaves only three hops — and the total is the same either way, so it is always allowed to start from the bigger one.

(b) Step 2: hold 5 in your head and hop three times: six (one), seven (two), eight (three).

Step 3: Ravi has 8 cars.

Step 4: check by counting the picture straight through: three filled dots and five open dots make eight ✓.

Watch for: 7. Fewer hops means fewer chances to drop one, which is the whole reason for starting big.

4. Find and fix: Jo has 5 shells and wants 9.

(a) Step 1: name the mistake. Jo counted the *words* she said, and the first word, “five”, was not a hop — it is where she started. The hops are six, seven, eight, nine, so she really took 4 hops.

(b) Step 2: each hop is one more shell, so the number of hops is the number of shells she still needs.

Step 3: Jo needs 4 more shells.

Step 4: check by adding it back: $5 + 4 = 9$ ✓. Jo’s answer would give $5 + 5 = 10$, which is too many.

Watch for: 5, the number of words said out loud. Counting words instead of hops is the same slip as counting the start number as a hop.

5. Find and fix: Ana, 2 marbles and 7 more.

(a) Step 1: Ana started at 2 and had seven hops to track, and one got lost. Starting at 7 leaves only two hops.

(b) Step 2: hold 7 in your head and hop twice: eight (one), nine (two).

Step 3: Ana has 9 marbles.

Step 4: check the other way round — starting at 2 and hopping seven times also lands on nine, but you have to say seven numbers without a slip ✓.

Watch for: 8. When the added number is big, always turn the problem around first.

6. Find and fix: Ben's two problems.

(a) Step 1: name the mistake. Both times Ben said his starting number as hop number one. That always makes the total exactly one too small, which is why both of his answers are off by the same amount.

Step 2: pencils — hold 6 in your head and hop four times: seven, eight, nine, ten.

Step 3: Ben has 10 pencils.

(b) Step 4: crayons — turn it around and start from the bigger number. Hold 6 in your head and hop twice: seven, eight.

Step 5: Ben has 8 crayons.

Step 6: check both by counting the picture dots straight through: the top row holds ten dots, the bottom row eight ✓.

Watch for: 9 and 7, each exactly one too small. When a child's answers are all one below the right answer, the fix is the start number, not more practice at counting.